

Abdullah NASSAR

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Education

M.Sc. in Meteorology and Climate Physics

Karlsruhe Institute of Technology (KIT)

Apr 2025 - Expected Apr 2027

Karlsruhe, Germany

- Thesis: Long-term variability and trends of compound precipitation and wind extreme events in Europe using a large ensemble of regional climate model simulations.

B.Sc. in Meteorological Engineering

Istanbul Technical University

Aug 2015 - Aug 2020

Istanbul, Turkey

- Graduated top of the class and earned placement on the Dean's High Honor List.
- Selected projects: Expected Drought Conditions in the Middle East; Disaster Management of Dust Storms; Drought Analysis in Australia.

Research Experience

Master's Thesis Research

Karlsruhe Institute of Technology (KIT)

2025 - Present

Karlsruhe, Germany

- Analyze LAERTES-EU, a regional climate model ensemble spanning approximately 12,500 simulated years, using Python, NetCDF, and Git.
- Identify concurrent precipitation and wind extremes and evaluate their long-term variability and trends across Europe.
- Apply MI and PSI statistical indices and construct a compoundness index to characterize multivariate extreme events.

Student Research Assistant (HiWi) - Cloud Physics

KIT Institute of Meteorology and Climate Research (IMK), Troposphere Research

Jan 2026 - Dec 2026

Karlsruhe, Germany

- Integrate global gridded FLASHMAP lightning data with CAMS aerosol fields and cloud-ice information in Python/Jupyter on the LAVANTE computing system.
- Evaluate relationships among lightning rate, aerosol loading, and cloud glaciation to support research on aerosol-cloud-lightning processes and climate-model representation.

Student Research Assistant (HiWi) - ACTRIS / INSEKT

KIT Institute of Meteorology and Climate Research (IMK), Atmospheric Aerosol Research

Feb 2026 - Mar 2027

Karlsruhe, Germany

- Maintain and extend an inherited Python GUI for INSEKT experiment control, improving stability and usability for laboratory work.
- Enable laboratory users to run the complete experimental workflow through an interface used during live experiments.
- Translate operational feedback into feature refinements and issue fixes while formal documentation remains in preparation.

Publications and Conference Contributions

Conference Abstract

2026

EGU General Assembly | Waman, D., Nassar, A., and Hoose, C. Regime-dependent Impacts of CCN and Cloud Glaciation on Global Lightning Activity. EGU26-3745.

doi.org/10.5194/egusphere-egu26-3745

- Co-authored a global analysis of lightning, aerosol, and cloud-microphysical data showing peak lightning activity at moderate CCN concentrations (400-600 cm⁻³) and a central role for cloud glaciation.

Technical Skills

Programming and development: Python, Git, Jupyter Notebook, Python GUI development; R, GrADS, C/C++, HTML/CSS

Data and computing: NetCDF, LAVANTE/HPC, statistical analysis, large-ensemble climate analysis, data visualization

Scientific datasets: LAERTES-EU, FLASHMAP, Copernicus Atmosphere Monitoring Service (CAMS)

Climate and atmospheric methods: Compound-event detection, MI/PSI and compoundness indices, aerosol-cloud-lightning analysis, regional climate modeling

Geospatial and hydrology: ArcGIS, watershed delineation, rating curves, flood estimation, hydropower applications

Research Interests

Focus areas: Machine learning for atmospheric and climate applications; compound precipitation and wind extremes; cloud and aerosol microphysics; lightning; regional climate modeling; energy meteorology

Teaching Experience

Mathematics Instructor (Part-Time, Online)

American Straight A Academy

Dec 2022 - Present

Online

- Deliver 30+ online courses, most comprising 40 instructional hours, for international students through ClassIn.
- Teach AP Calculus AB/BC, Algebra, and Precalculus, adapting explanations and practice to varied academic backgrounds.

High School Mathematics Teacher

Al-Jazari International Schools

Sep 2022 - Aug 2023

Istanbul, Turkey

- Taught grades 9, 11, and 12 in AP Calculus AB/BC, Precalculus, and Edmentum Integrated Mathematics.
- Prepared annual and weekly curriculum plans, selected learning materials, and contributed to the Exam and Report Card Committee.

Mathematics and Physics Teacher

Al-Aqsa International Schools

Nov 2021 - Aug 2022

Istanbul, Turkey

- Taught mathematics to grades 10-12 and physics to grades 9-11.
- Prepared annual and weekly plans and classroom materials for McGraw Hill Mathematics, Physics, and Precalculus.

Volunteer English Instructor

GoCamp NGO

Jun 2018 - Jul 2018

Lviv, Ukraine

- Guided approximately 30 children through a three-week summer program, teaching groups of 10-15 learners.

Applied Hydrology Experience

Hydrology Intern

TMMOB Chamber of Meteorological Engineers

Aug 2019 - Sep 2019

Ankara, Turkey

- Completed a 19-day internship covering hydrometric monitoring, rating curves, watershed mapping, water potential calculations, flood estimation, dam operations, and hydropower feasibility.
- Processed one month of station D15A195 observations and identified a 0.84 million m³ difference between daily and interpolated monthly discharge estimates.
- Applied the Mockus method and an SCS dimensionless unit hydrograph to a Nesetiy Dam exercise, deriving a peak unit discharge of 5.425 m³/s/mm.
- Used ArcGIS and topographic maps to delineate rainfall areas, prepare elevation maps, and document the analyses in a signed internship report.

Additional Professional Experience

Fundraising Assistant (Remote)

Resonate Yemen Foundation

Dec 2023 - Present

Sana'a, Yemen

- Contribute to 22+ funding-call applications across 2025 and 2026 while maintaining donor and opportunity tracking.
- Screen donor requirements, draft concept notes and proposals, support institutional registrations, and coordinate submission correspondence.

Contracts and Tenders Specialist

Wedge for Architectural Contracting

Oct 2023 - Sep 2024

Sana'a, Yemen

- Interpreted tender requirements, specifications, and risk considerations for a newly established contracting company.
- Coordinated submissions with internal contributors and managed clarifications, records, and post-submission negotiations.

Operations Manager

Hope Investment Enerji Sanayi ve Ticaret Limited Sirketi

Jun 2022 - Sep 2023
Istanbul, Turkey

- Oversaw a three-vehicle family rental fleet serving international customers, including vehicle readiness, KABIS registrations, traffic-system records, and invoicing.

Leadership and Community Service

Cultural and Academic Administrator

Yemeni Students Association - Turkey, Istanbul

Sep 2019 - Oct 2020
Istanbul, Turkey

- Served a community of 1,000+ Yemeni students through enrollment coordination, academic guidance, and liaison with the Yemeni Embassy.
- Coordinated 30+ training, academic, and community events during the appointment.

Vice President, 3rd General Conference

Yemeni Students Association - Turkey

Dec 2019 - Feb 2020
Kusadasi, Turkey

- Co-led a biennial conference involving 20+ student unions and 200+ participants.
- Supervised conference delivery, the election process, and the transition of leadership.

Awards and Scholarships

Top Performing Student in Meteorological Engineering

Istanbul Technical University

2020

Dean's High Honor List

Istanbul Technical University

2020

Yemen Ministry of Higher Education Scholarship

Awarded to high-performing secondary-school graduates

2015

Professional Development

Civic Engagement Leadership Institute - MEPI Student Leaders Program

Roger Williams University | Six-week program

2014

Seattle Leadership Program - Middle East Partnership Initiative

World Affairs Council

2014

Languages

Arabic: Advanced | **English:** Advanced (IELTS/TOEFL) | **Turkish:** C1 | **German:** Basic